SAMPLE

Chaos Theory

The Butterfly Effect and Deterministic Chaos

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Introduction

Chaos theory is a branch of mathematics and an interdisciplinary area of scientific study. It focuses on underlying patterns and deterministic laws of dynamical systems that are highly sensitive to initial conditions. These were once thought to have completely random states of disorder and irregularities. The theory states that within the apparent randomness of chaotic complex systems, there are underlying patterns, interconnection, constant feedback loops, repetition, self-similarity, fractals and self-organization.

The butterfly effect, an underlying principle of chaos, describes how a small change in one state of a deterministic nonlinear system can result in large differences in a later state (meaning there is sensitive dependence on initial conditions). A metaphor for this behavior is that a butterfly flapping its wings in Brazil can cause or prevent a tornado in Texas.

Small differences in initial conditions, such as those due to errors in measurements or due to rounding errors in numerical computation, can yield widely diverging outcomes for such dynamic systems, rendering long-term prediction of their behavior impossible in general. This can happen even though these systems are deterministic, meaning that their future behavior follows a unique evolution and is fully determined by their initial conditions, with no random elements involved.

In other words, despite the deterministic nature of these systems, this does not make them predictable. This behavior is known as deterministic chaos, or simply chaos. The theory was summarized by Edward Lorenz as: Chaos: When the present determines the future but the approximate present does not approximately determine the future. Chaotic behavior exists in many natural systems, including fluid flow, heartbeat irregularities, weather and climate. It also occurs spontaneously in some systems with artificial components, such as road traffic.

Chapter 1: The Discovery of Chaos Theory

Chaos theory concerns deterministic systems which are predictable for some amount of time and then appear to become random.